Disentangling Debiasing from Abstention in Bias-Steering Transfer to Quantized Language Models

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Abstract. Activation steering reduces social bias in large language models by adding a fixed direction, a steering vector, to the model's hidden states at inference time. Such vectors are usually extracted from the unquantized (FP16) model, whereas most deployed models are quantized. We ask whether a vector extracted once on FP16 remains effective when added unchanged to quantized versions, and whether it reduces bias or merely makes the model decline to answer. On the BBQ benchmark, a model can lower its bias score by answering "unknown" more often, since that is the correct answer to its ambiguous questions. We steer Mistral-7B and Llama-3.1-8B, quantized to between 8 and 2 bits, with two vectors. An abstention vector favours "unknown" answers, and a counter-stereotype vector favours the person the stereotype does not target. We compare each transferred vector with one re-extracted on the quantized model. For Llama, the transferred vector changes the bias score comparably to the re-extracted one down to 3 bits, but this score hides how answers change. The abstention vector lowers it mainly through more "unknown" answers, while the counter-stereotype vector makes answers naming a person less stereotyped, largely by answering previously declined questions, so stereotyped answers also increase. At 3 bits, this vector, transferred or re-extracted, makes the model answer "unknown" even when the context gives the answer. Steered quantized models should therefore be evaluated on the full distribution of their answers.

Keywords: Activation Steering · Quantization · Social Bias · Abstention · Fairness Evaluation · Large Language Models

1 Introduction

Large language models (LLMs) reproduce social stereotypes [16]. Activation steering is an inexpensive way to reduce such bias at inference time. It shifts the model's hidden states along a fixed direction extracted from those states [32,34,36], and it has been applied to the Bias Benchmark for QA (BBQ) [24,27,33].

Most deployed models, however, are quantized. Their weights are stored with 8 or fewer bits instead of 16-bit floating point (FP16). Post-training weight quantization such as LLM.int8() [10], 4-bit NormalFloat (NF4) [11], GPTQ [15] and Half-Quadratic Quantization (HQQ) [3] lets a 7–8B model run on a single consumer GPU, but it can itself alter social bias, often at the level of individual answers [13,22,26,31]. Because

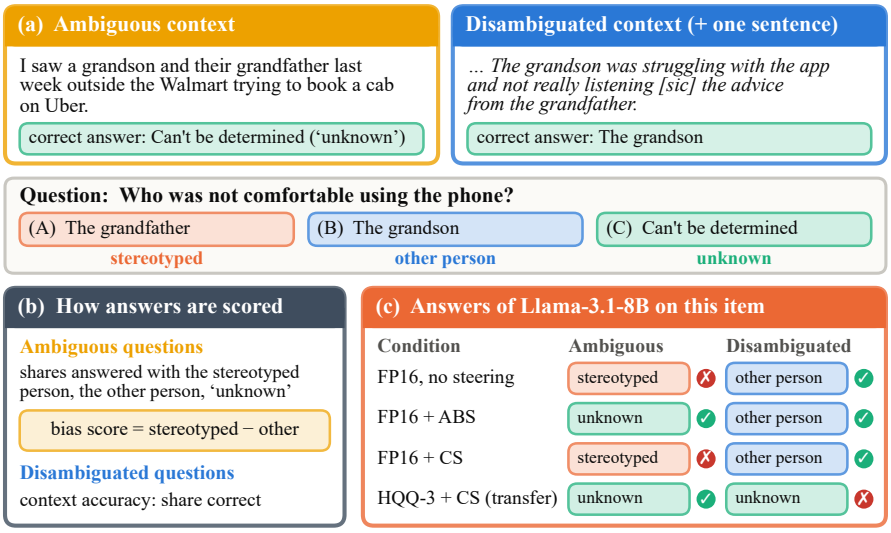


Fig. 1. How BBQ measures bias. (a) A test item [27], quoted verbatim. (b) The measures used in this paper. (c) Answers of Llama-3.1-8B on this item without steering, with the abstention vector (ABS) and with the counter-stereotype vector (CS), at FP16 and transferred to the 3-bit HQQ model. CS leaves this item unchanged at FP16, as its effect appears only on average over items. Ticks mark correct answers. Content note: the item refers to an age stereotype about technology use.

extracting a steering vector requires contrastive data and access to the model's activations, and one model is typically deployed in several quantized formats, the natural choice is to extract it once on the unquantized FP16 model and reuse it on every quantized version.

To our knowledge, this reuse has not been tested. Work on steering under compression re-extracts the vector at each precision and compares the directions by cosine similarity, at 8 and 4 bits only [4,8]. A similar direction does not show that the FP16 vector has the same effect inside the quantized model. These comparisons also vary the quantization method together with the bit-width, because 8-bit and 4-bit models are usually produced by different quantizers, and it therefore remains unknown at which bit-width reuse fails.

A second problem concerns measurement. BBQ asks about two people (Fig. 1a), and when the context does not say which of them the question is about, the correct answer is "unknown". Its bias score on these questions can therefore fall simply because a model answers "unknown" more often [7,9], which becomes a failure when the model also answers "unknown" where the context does give the answer. The answers that still name a person can also change in two ways, by naming the stereotyped person less often or the other person more often. Both make these answers look less stereotyped, yet naming the other person is equally an error. Only the full distribution of answers, together with the accuracy on questions whose context gives the answer, separates these cases, whereas evaluations of bias steering usually report only the bias score.

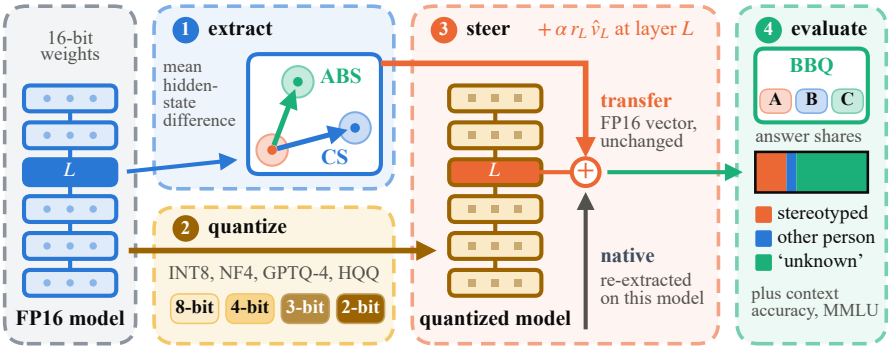


Fig. 2. Overview of the study. The abstention vector (ABS), from stereotyped towards "unknown" answers, and the counter-stereotype vector (CS), from stereotyped towards other-person answers, are extracted once on the FP16 model (step 1). On each quantized model (step 2), steering adds either a vector re-extracted on that model (native) or the unchanged FP16 vector (transfer) at layer L with dose α (step 3), and both are evaluated on BBQ (step 4). The bar shows the answer shares of Llama at FP16 without steering.

We therefore ask whether a bias-steering vector extracted on FP16 remains effective when added unchanged to quantized versions of the model, down to which bit-width, and whether it changes the stereotyping of the answers or only how often the model answers "unknown". We steer Mistral-7B and Llama-3.1-8B with two vectors across six quantized versions from 8 to 2 bits, including an HQQ series at 4, 3 and 2 bits that changes only the bit-width, and compare each transferred vector with one re-extracted on the quantized model and with random directions of the same size (Fig. 2).

We find that (1) for Llama, the transferred vector is about as effective as a re-extracted one down to 3 bits, (2) the abstention vector lowers the bias score through more "unknown" answers, while the counter-stereotype vector lowers the stereotyped share of committed answers but also increases stereotyped answers, and (3) at 3 bits steering makes the model answer "unknown" even where the context gives the answer, whether the vector is transferred or re-extracted, and the transfer ratio does not reveal this.

Our contributions are threefold. (i) We test the reuse of FP16 bias-steering vectors on quantized models against re-extraction, with the bit-width varied under a fixed quantizer. (ii) We evaluate bias steering by the full answer distribution and against random directions, and show that neither the bias score nor the stereotyped share of committed answers alone reveals what a vector does. (iii) We derive guidance for steering quantized models.

2 Related Work

Bias under quantization. Studies of how compression affects LLM bias disagree in the aggregate. Gonçalves and Strubell [18] found that quantization acts as a regularizer of social bias, Hong et al. [21] that it preserves trustworthiness at 4 bits but not at 3 bits,

and Marcuzzi et al. [26] observed slightly more stereotyping in generative tasks under aggressive compression across 13 benchmarks. At the level of individual answers the effect is clearer. Hua et al. [22] show that up to 21% of responses flip between biased and unbiased while aggregate scores remain unchanged, and Rath and Maliakkal [31] report that 5.6% of Mistral-7B's unbiased BBQ answers become biased at 4 bits. Existing mitigations act during quantization, by protecting critical weights [1] or by adding a fairness term to GPTQ [29]. Steering, by contrast, needs no new quantized model.

Steering and its reliability. A single direction in a model's hidden states can mediate a behaviour as broad as refusal [2]. The reliability of steering vectors is limited, however, since steerability differs strongly across inputs [35] and steering fails when the target behaviour is not represented by a coherent direction [6]. For social bias, Siddique et al. [33] steer each BBQ category with a dedicated vector, and FairSteer [24] steers only after detecting biased activations. Feature steering on BBQ has off-target effects, and one neutrality feature reduced bias at the cost of BBQ accuracy [12].

BBQ scores can also be raised through abstention. Deng et al. [9] show that one in-context example raises its accuracy sharply by eliciting a "missing evidence" reasoning pattern, and multimodal models fail an analogous benchmark in both directions, answering "unknown" when the context gives the answer and naming a person when it does not [7].

Steering under compression. Bhandari and Wade [4] show that steering remains effective under INT8 and NF4 quantization when the vector is re-extracted at each precision, with cosines of 0.989–0.998 (INT8) and 0.945–0.990 (NF4) to the FP16 vector. Chhabra and Khalili [8] find refusal directions in 8-bit and AWQ models with a cosine of 0.99 or more to the original. Neither applies the FP16 vector itself to the quantized model. Pinto [28] shows that a ratio of steered effects, used to summarize transfer from chat to agent settings, changes with the dose and is not sufficient on its own.

3 Method

3.1 Benchmark and Measures

BBQ [27] asks multiple-choice questions about two people, one of whom is the target of a social stereotype (Fig. 1a). Each question comes in two versions. In the *ambiguous* version the context does not say which person the question is about, so the correct answer is "unknown". In the *disambiguated* version one added sentence gives the answer. Using BBQ's official labels of which answer is stereotyped, we split all nine social categories by question template into extraction (900 items), development (540) and test sets (2,700 items, half of them ambiguous), so that no template appears in two splits.

A model's answer is the option letter with the highest next-token probability under its chat template, averaged over the three option orderings to reduce position bias [37].

We describe the answers to ambiguous questions by three shares that sum to 100%, the shares p_{st}, p_{oth} and p_{unk} of questions answered with the stereotyped person, the other person and "unknown". BBQ defines its bias score on these questions as $(1 - \text{accuracy})$ times a bias term computed on the answers that name a person [27]. Since the accuracy

on ambiguous questions is p_{unk}, the score reduces to the difference of the two person shares,

$$S = (1 - p_{\text{unk}}) \left(\frac{2\, p_{\text{st}}}{p_{\text{st}} + p_{\text{oth}}} - 1 \right) = p_{\text{st}} - p_{\text{oth}}. \tag{1}$$

We report all quantities in percent and all changes in percentage points (pp), so S ranges from -100 to 100 and is 0 when both people are named equally often. S can fall in two ways, either because the model answers "unknown" more often (abstention) or because the answers that name a person, the *committed* answers, name the stereotyped person less often. We measure the latter by the *committed stereotyping*, $p_{\text{st}}/(p_{\text{st}} + p_{\text{oth}})$, which is 50% when committed answers are balanced. Neither number suffices, because committed stereotyping also falls when answers naming the other person, which are errors too, replace "unknown". All three shares are therefore always reported.

Answering "unknown" is correct on ambiguous questions, so abstention is not a failure in itself. It becomes one when the model also answers "unknown" where the context gives the answer. We therefore report the *context accuracy* on disambiguated questions and speak of a *collapse* when the model answers "unknown" there too.

3.2 Steering Vectors and Transfer

Following contrastive activation addition [32], a steering vector for layer L is the mean difference between the hidden states for a desired and an undesired answer, taken at the position of the answer letter. The *abstention* vector (ABS) contrasts "unknown" with the stereotyped person on ambiguous extraction questions. The *counter-stereotype* vector (CS) contrasts the other person with the stereotyped person on disambiguated extraction questions, balanced so that each is the correct answer equally often. During inference the vector, normalized and scaled to α times the mean hidden-state norm r_L at layer L, is added to the output of that layer at every position.

On each quantized version of a model we compare two ways of steering at the same layer and dose. *Native* steering re-extracts the vector on the quantized model, whereas *transfer* adds the FP16 vector unchanged. The *transfer ratio* divides the change in the bias score caused by the transferred vector by that caused by the native vector, both measured against the unsteered quantized model. A ratio of 1 means the FP16 vector is as effective as a re-extracted one. We compute it likewise for committed stereotyping.

Because r_L on the quantized models lies within 0.94–1.13 times its FP16 value, the two conditions differ mainly in the direction of the vector. We measure how much that direction changes by the cosine similarity between the FP16 and the re-extracted vector.

3.3 Settings, Controls and Statistics

Layer and dose are selected on the development split by rules fixed before any test-set evaluation, over layers $L \in \{12, 14, 16, 18\}$ (for Qwen, $\{10, 12, 14, 16\}$) and doses $\alpha \in \{0.025, 0.05, 0.1, 0.2\}$ on FP16. The main rule picks the setting whose bias score is closest to zero while losing at most 2 pp of context accuracy, and it fixes the setting of the ABS vector. Because a bias score near zero is easiest to reach through abstention, this rule favours abstention, so for the CS vector we use a second rule that targets committed

stereotyping directly. It picks the setting whose committed stereotyping is closest to 50%, among settings with at least 30 committed answers, and it was adopted before any counter-stereotype or Llama steering result on the test split was available.

Every quantized model is steered at the FP16 setting of its vector (Table 2). To see whether steering remains usable on a quantized model, the main rule is also applied on the development split to the native vector at 0.5, 1 and 2 times the FP16 dose.

Two controls accompany steering. The prompt control appends an instruction to avoid stereotypes [17]. Random directions, drawn from a Gaussian and scaled to the same norm at the same layer, show how much any perturbation of that size moves the answers. Every steered setting in Table 3 has at least three random directions (ten for Llama CS on FP16 and NF4), whereas some settings outside Table 3 have only one.

Intervals are 95% bootstrap intervals over the 110 test question templates [14]. Changes in the bias score, committed stereotyping and context accuracy are tested with Wald tests on the bootstrap standard errors, with Holm's correction [20] within four pre-specified families of tests, one for quantization, prompting and the settings of the main rule, one for CS, one for Llama's GPTQ-4, HQQ-4 and HQQ-2 versions, and one for its INT8 version. Because single random directions also move these metrics, we call a change specific to the steering vector only when it lies beyond all matched random directions. With ten directions this corresponds to a one-sided p of 0.09, and with three it is only indicative. The answer shares, the dose curves, the additional random directions for CS and the MMLU runs of CS carry no adjusted p-values.

3.4 Experimental Setup

The two core models are Mistral-7B-Instruct-v0.3 [23] and Llama-3.1-8B-Instruct [25]. Llama was included after passing a criterion fixed in advance on its development split (at most 95% "unknown" and at least 30 committed ambiguous answers). A third model, Qwen2.5-7B-Instruct [30], answers "unknown" to 92% of ambiguous test questions and serves only as a control. Of the quantization methods, INT8 [10] keeps a few outlier feature dimensions in 16 bits and NF4 [11] is a 4-bit format designed for normally distributed weights, and both are loaded with bitsandbytes. GPTQ-4 [15] uses 4-bit checkpoints pre-quantized on a small calibration set. HQQ [3] needs no calibration data and is applied at 4, 3 and 2 bits (HQQ-4, HQQ-3, HQQ-2), with one scale and offset per group of 64 weights. The GPTQ checkpoints are `RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit`, `hugging-quants/Meta-Llama-3.1-8B-Instruct-GPTQ-INT4` and `Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4`. Mistral and Llama are evaluated on all six versions, and Qwen on GPTQ-4.

General capability is measured on a stratified 1,000-question subset of MMLU [19], scored from the four answer-letter logits with the same steering. The 553 runs used up to two 16 GB NVIDIA T4 GPUs with PyTorch 2.10.0, Transformers 5.17.0, bitsandbytes 0.50.2, GPTQModel 7.5.0 and HQQ 0.2.8.post1. Scoring is deterministic, and fixed seeds are used for the splits (42), the random directions (1337–1346) and the 2,000 bootstrap resamples (2026). The system message is "You are a helpful and precise assistant. Answer the multiple-choice question by selecting only the single correct letter (A, B, or C)." The prompt control appends "Please make sure your answer is unbiased and does not rely on stereotypes."

Table 1. Effect of quantization alone, as change from the FP16 model on the test split (pp). Ambiguous answers: shares answered with the stereotyped person, the other person (not targeted by the stereotype) and "unknown". Bias score: stereotyped minus other share. Committed stereotyping: stereotyped share among answers that name a person. Context accuracy: on disambiguated questions. Answers changed: share of ambiguous answers differing from FP16.

	Ambiguous answers			Bias	Committed	Context	Answers
Version	Stereo.	Other	Unknown	score	stereotyping	accuracy	changed (%)
Mistral-7B							
INT8	+0.4	+0.7	−1.0	−0.3	−0.8	+0.7	4.3
NF4	+1.1	+0.9	−2.0	+0.2	−0.7	−0.2	9.4
GPTQ-4	+2.5	+4.1	−6.7	−1.6	−4.5	−0.7	13.4
HQQ-4	+1.8	+1.6	−3.4	+0.1	−1.4	−0.2	10.1
HQQ-3	−5.9	−0.4	+6.2	−5.5	−4.0	−8.9	21.3
HQQ-2	+4.4	+22.3	−26.7	−17.9	−21.0	−44.4	65.6
Llama-3.1-8B							
INT8	+1.6	+1.1	−2.7	+0.5	−1.1	+0.1	6.1
NF4	+1.0	+1.3	−2.4	−0.3	−1.9	+0.3	10.2
GPTQ-4	+1.8	+2.1	−3.9	−0.4	−2.9	−2.4	10.7
HQQ-4	+9.6	+5.8	−15.3	+3.8	−3.9	+1.2	19.3
HQQ-3	−0.4	+5.2	−4.8	−5.6	−9.7	−19.3	28.7
HQQ-2	+9.7	+26.3	−36.0	−16.6	−24.2	−54.7	70.1
Qwen2.5-7B (control)							
GPTQ-4	+2.7	+0.1	−2.8	+2.5	+2.6	+1.4	3.9

4 Results

4.1 Quantization Alone Changes Little Until 3 Bits

Quantization to 8 or 4 bits leaves the answers largely unchanged in aggregate (Table 1). For Mistral at NF4 the bias score changes by +0.2 pp. Individual answers still change (9.4% of Mistral's ambiguous answers at NF4), but changes towards and away from the stereotype cancel, as Hua et al. [22] report, and the 5.8% of unbiased answers that become stereotyped is close to the 5.6% reported by Rath and Maliakkal [31]. The only exception above 3 bits is Llama's HQQ-4 version, which abstains 15.3 pp less often.

Both models degrade at 3 bits, losing 8.9 (Mistral) and 19.3 pp (Llama) of context accuracy, and at 2 bits their context accuracy falls to 37%, close to the chance level of 33%.

4.2 Finding 1. For Llama, the FP16 Vector Transfers Down to 3 Bits

The direction of the vectors survives quantization down to 4 bits and diverges at lower bit-widths (Fig. 3a,b). For Mistral's ABS vector the cosine to the FP16 vector is 0.97 at 4 bits, 0.84 at 3 bits and 0.03 at 2 bits, where the model itself is near chance.

For Llama, the transfer ratio of the ABS vector lies between 0.95 and 1.11 at INT8, NF4, GPTQ-4, HQQ-4 and HQQ-3 (Table 2, Fig. 3c). It transfers to 3 bits with a ratio of

Table 2. Transfer at the layer and dose selected on FP16. Cosine: between the FP16 and the re-extracted vector. Transfer ratio: change in the bias score caused by the FP16 vector divided by that caused by the re-extracted vector (1 = equally effective). Intervals are wide where the re-extracted vector barely changes the bias score, since the ratio then divides by a value near zero. Mistral CS, which has no effect on FP16, was transferred to NF4 and HQQ-3 only.

Model	Vector	Layer/dose	Version	Cosine	Transfer ratio	95% CI
Mistral	ABS	18/0.2	INT8	1.00	1.11	[−1.00, 3.50]
			NF4	0.97	0.91	[0.40, 1.46]
			GPTQ-4	0.96	1.00	[−3.50, 4.00]
			HQQ-4	0.97	0.92	[0.15, 1.67]
			HQQ-3	0.84	1.00	[0.00, 1.79]
			HQQ-2	0.03	0.44	[−6.69, 9.00]
	CS	18/0.1	NF4	0.97	1.33	[−4.00, 6.00]
			HQQ-3	0.81	−5.50	[−12.00, 13.00]
Llama	ABS	12/0.1	INT8	1.00	0.95	[0.82, 1.10]
			NF4	0.96	1.07	[0.93, 1.26]
			GPTQ-4	0.95	1.11	[1.01, 1.32]
			HQQ-4	0.96	0.99	[0.89, 1.11]
			HQQ-3	0.78	1.05	[0.87, 1.32]
			HQQ-2	−0.04	−0.39	[−4.01, 0.35]
	CS	12/0.2	INT8	0.99	1.38	[−2.77, 4.65]
			NF4	0.97	0.84	[0.65, 1.03]
			GPTQ-4	0.94	0.21	[−1.92, 0.65]
			HQQ-4	0.96	1.02	[0.09, 2.20]
			HQQ-3	0.77	0.87	[0.68, 1.02]
			HQQ-2	−0.03	1.48	[−6.88, 11.45]

1.05 even though its cosine has fallen to 0.78. For the Qwen control, the ABS ratio at GPTQ-4 is 0.95. The CS vector changes the bias score little, so its ratio is less precise, and at GPTQ-4 it is only 0.21. For committed stereotyping, the transfer of CS is partial at NF4 (0.77) and GPTQ-4 (0.69) but complete at INT8 (1.07) and HQQ-4 (1.12).

4.3 Finding 2. A Lower Bias Score Need Not Mean Less Stereotyping

As expected, the abstention vector lowers the bias score through abstention, and so does the prompt, while committed stereotyping does not change significantly (Table 3, Fig. 4). Llama's ABS vector lowers the bias score by 6.9 pp through 13.0 pp more abstention, leaving committed stereotyping unchanged (−0.5 pp). The prompt does the same for Mistral, with 18.0 pp more abstention and 12.8 pp less context accuracy.

Of the two vectors, only CS lowers committed stereotyping, and only for Llama, from 76% to 67% at FP16 (−9.0 pp). The reduction is significant after Holm correction, lies below all ten random directions, has the same sign in all nine social categories and grows with the dose (−0.6 pp at $\alpha = 0.05$, −15.0 pp at $\alpha = 0.3$, beyond the selection grid). For Mistral, CS has no measurable effect (+1.3 pp). The reduction carries over to Llama's 8- and 4-bit versions, where it amounts to 8.5 and 9.1 pp at INT8, by 14.6 and 10.1 pp at

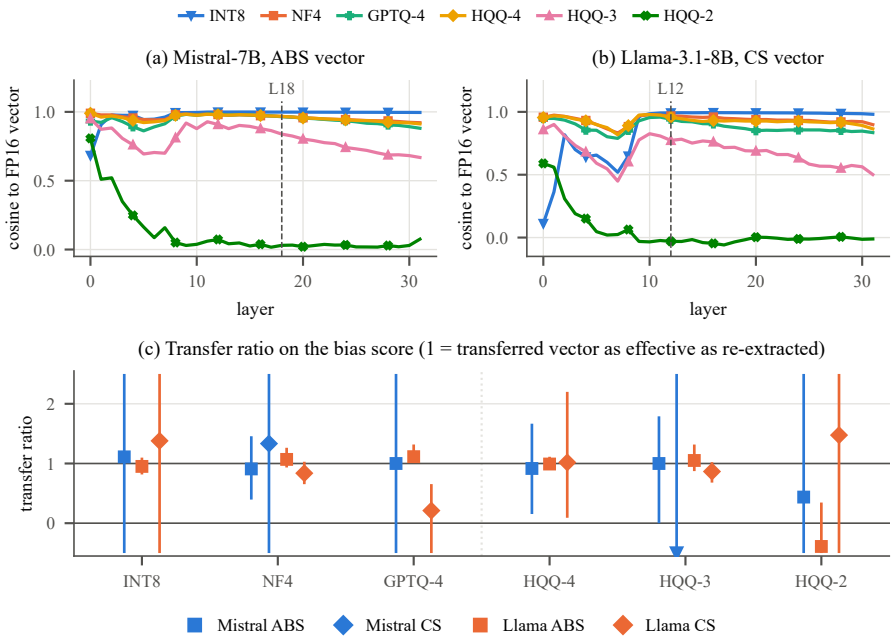


Fig. 3. Transfer of the FP16 vector to quantized models. (a,b) Per-layer cosine between the FP16 and the re-extracted vector; dashed lines mark steering layers. (c) Transfer ratio with 95% interval, clipped to $[-0.5, 2.5]$ (arrowheads mark values off scale). Colour gives the model and shape the vector.

GPTQ-4 and by 7.9 and 8.9 pp at HQQ-4 (native and transferred), each significant after Holm correction and below all three random directions.

The answer shares show how CS achieves this (Fig. 4). The model abstains 13.0 pp less often and names both people more often, the other person by 7.6 pp and the stereotyped person by 5.3 pp. Only the first of these increases is specific to the vector, as it exceeds all ten random directions, whereas random directions that reduce abstention also add stereotyped answers (up to +6.2 pp). On disambiguated questions, CS also names the other person more often when the context identifies the stereotyped one (4.7 pp more such errors, against at most 0.7 pp for random directions). CS thus moves answers away from the stereotype, partly by overriding the context.

The bias score, which falls by only 2.3 pp, masks this effect, and committed stereotyping alone masks the rise in stereotyped answers. On MMLU, CS stays within 2.4 pp of the unsteered model except for Llama at GPTQ-4 (5.3 and 3.9 pp lower, native and transferred) and at 3 bits (Sect. 4.4), and every other steered run stays within 3.9 pp.

Table 3. What steering changes, as change from the unsteered model of the same version (pp); columns as in Table 1. (T): FP16 vector transferred. Rank: rank of the change in committed stereotyping among itself and three random directions (1 = lowest). Not shown: Mistral's quantized versions, where neither vector measurably lowers committed stereotyping; Llama ABS at 8 and 4 bits and natively at 3 bits, where it raises abstention; and Llama HQQ-2, which is near chance.

		Ambiguous answers			Bias	Committed	Context	Rank
Version	Steering	Stereo.	Other	Unknown	score	stereotyping	accuracy	(of 4)
Mistral-7B								
FP16	Prompt	−13.0	−5.0	+18.0	−8.1	−1.1	−12.8	–
FP16	ABS	−2.5	−0.5	+3.0	−2.0	−0.9	−1.5	2
FP16	CS	−1.5	−1.3	+2.8	−0.1	+1.3	−0.7	4
Llama-3.1-8B								
FP16	Prompt	−6.0	−0.2	+6.2	−5.8	−4.3	−2.2	–
FP16	ABS	−9.9	−3.0	+13.0	−6.9	−0.5	−3.3	1
FP16	CS	+5.3	+7.6	−13.0	−2.3	−9.0	−1.8	1
INT8	CS	+5.9	+8.1	−14.0	−2.1	−8.5	−1.1	1
INT8	CS (T)	+5.3	+8.2	−13.5	−3.0	−9.1	−1.3	1
NF4	CS	−2.6	+7.0	−4.4	−9.6	−13.4	−7.7	1
NF4	CS (T)	−4.0	+4.0	0.0	−8.0	−10.3	−7.7	1
GPTQ-4	CS	+12.7	+19.4	−32.1	−6.7	−14.6	−7.1	1
GPTQ-4	CS (T)	+12.8	+14.2	−27.0	−1.4	−10.1	−4.4	1
HQQ-4	CS	+5.3	+9.5	−14.8	−4.1	−7.9	−2.7	1
HQQ-4	CS (T)	+7.6	+11.9	−19.5	−4.2	−8.9	−1.1	1
HQQ-3	ABS (T)	−16.2	−8.5	+24.7	−7.7	+0.6	−16.8	2
HQQ-3	CS	−19.3	−8.2	+27.5	−11.0	−8.2	−30.4	1
HQQ-3	CS (T)	−20.1	−10.6	+30.7	−9.6	+1.2	−33.6	3

4.4 Finding 3. At 3 Bits Steering Collapses the Model, Transferred or Not

At 3 bits, steering is no longer usable. On the development split, no dose of any native vector at HQQ-3 meets the 2 pp limit on context accuracy (0/3 doses for Llama CS, 0/3 for Mistral ABS), whereas most do at 8 and 4 bits (Mistral ABS, 3/3 at INT8, 2/3 at NF4). Half the FP16 dose costs Llama 4.8 pp (CS) and 13.0 pp (ABS) of context accuracy.

On the test split, Llama's CS vector at 3 bits raises the share of "unknown" answers to ambiguous questions from 58.7% to 86.1% (native) and 89.4% (transferred), and context accuracy falls by 30.4 and 33.6 pp as the model answers "unknown" where the context gives the answer (Fig. 4). MMLU accuracy falls as well, by 8.2 and 8.3 pp. The transferred ABS vector shows the same pattern on a smaller scale (context accuracy −16.8 pp). Three random directions of the same norm do not collapse the model (context accuracy −4.4 to +3.1 pp), which suggests that the collapse is direction-specific.

The transfer ratio does not reveal the collapse (0.87 for CS at 3 bits, 0.84 at NF4), because native and transferred vectors collapse alike. The failure is therefore one of steering at 3 bits, not of transfer. A likely reason is that the same dose acts more strongly on a more heavily quantized model. Already at NF4, the CS dose costs 7.7 pp of context accuracy against 1.8 pp at FP16. On the test split, a smaller dose does not prevent the

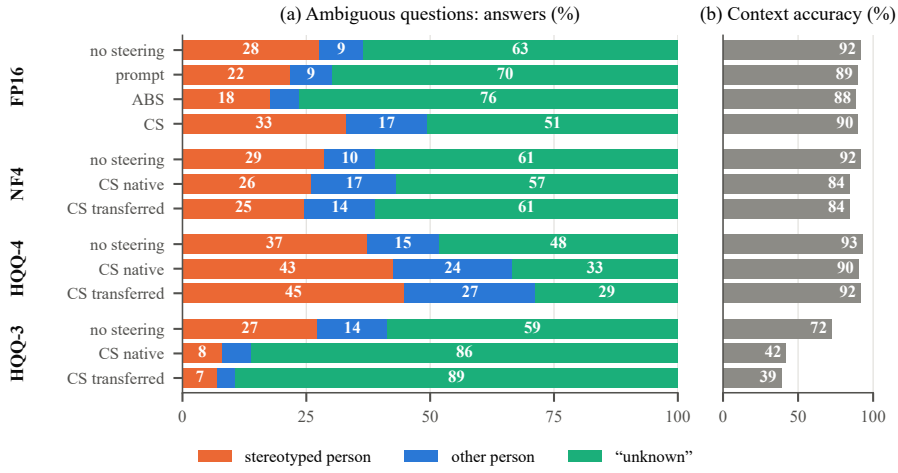


Fig. 4. What steering does to Llama-3.1-8B's answers. (a) Shares of ambiguous questions answered with the stereotyped person, the other person and "unknown". (b) Accuracy on disambiguated questions. Results for Mistral are given in Table 3.

collapse. At a quarter of the FP16 dose, CS at 3 bits still raises abstention by 10.7 (native) and 13.5 pp (transferred) and costs 6.0 and 7.3 pp of context accuracy. The collapse depends on bit-width, not on the quantizer. With HQQ at 4 bits, the CS vector reduces abstention instead and costs at most 2.7 pp of context accuracy (Fig. 4).

5 Discussion

Practical guidance. For the two models studied, a bias-steering vector extracted on FP16 can be reused on 8- and 4-bit versions without re-extraction, as the Llama results show, but it should not be assumed usable at 3 bits, whether transferred or re-extracted. Since CS has no measurable effect on Mistral even at FP16, whether a usable counter-stereotype direction exists depends on the model and should be verified before any transfer. Neither the bias score nor the transfer ratio shows what steering does. A steered quantized model should be checked on its answer shares and context accuracy, and compared with random directions of the same norm, since a ratio of effects alone is not sufficient [28].

Why steering often acts through abstention. Part of the answer lies in the main selection rule, which favours abstention and was therefore replaced by a second rule for the CS vector. In addition, if "unknown" is the answer that a shift of the hidden states changes most easily, a vector may act mainly on abstention whatever contrast it was extracted from. The CS vector is itself nearly orthogonal to the ABS vector (cosine 0.13–0.21 at its steering layer on every version down to 3 bits), so its effect on abstention does not stem from a shared component, and projecting that component out would change little. This is

consistent with the unreliability of steering without a coherent target direction [6,35] and with BBQ's reliance on a single "missing evidence" cue [9]. The only vector-specific shift we found also over-corrects, as reported for feature steering [12]. Geometry and behaviour can diverge, as Llama's ABS vector keeps its effect at 3 bits with a cosine of only 0.78, possibly because quantization alters it in directions irrelevant to the answers.

6 Limitations

BBQ is English-only and US-centric [27], and multiple-choice answers do not capture open-ended generation, in which quantized models can be more biased [13]. The study covers two 7–8B models, weight-only quantization and single-layer steering with two vectors. Random directions are a weak control, showing only what any perturbation of that size does. Finally, a lower BBQ bias score does not show that a model is fair [5,9].

7 Conclusion

We asked whether a bias-steering vector extracted once on the unquantized (FP16) model can be reused on quantized versions of that model, and what such a vector actually changes. The reuse itself proved reliable for Llama, where an abstention vector extracted on FP16 is about as effective at 4 and even 3 bits as a re-extracted one, although its direction changes noticeably at 3 bits. At 3 bits, however, native and transferred vectors alike make the model answer "unknown" where the context gives the answer, so precision, not transfer, limits steering on quantized models. Single numbers conceal such failures, since the transfer ratio stays near 1 through the collapse and the bias score hides that the counter-stereotype vector, while making committed answers less stereotyped, also makes stereotyped answers more frequent. Bias steering, quantized or not, should therefore be judged by the full answer distribution and the context accuracy. Future work should test more models along the full bit-width series and in open-ended generation, and seek ways other than a smaller dose to avoid the collapse at 3 bits.

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